RE030 – Additive Risk Score: Weighting Criteria Development

Executive Summary

The Additive Risk Score (ARS) is proposed as a composite measure integrating Symptom Severity, Frequency in Products, and Cumulative Exposure Risk. Default weights are Severity 50%, Frequency 30%, and Exposure 20%. The ARS is transparent, consumer-friendly, and supported by established frameworks. It produces numeric scores and user-facing bands (Low, Moderate, High, Very High). It also provides rationales for each additive, ensuring explainability and trust.

1. Objectives and Scope

The ARS aims to provide consumers, regulators, and developers with a balanced view of additive safety. Its objectives include:

- Helping consumers quickly interpret additive risks while shopping.
- Supporting regulators and product teams in identifying high-risk formulations.
- Accounting for immediate, acute effects such as allergic reactions.
- Considering long-term cumulative exposure and chronic toxicity.
- Highlighting prevalence of additives to reflect real-world likelihood of exposure.
- Offering safeguards for red-flag additives that present disproportionate risks.
- Allowing cohort-specific adjustments, so users with conditions like celiac disease or soy allergy can receive personalised risk outputs.
- Providing transparency by publishing underlying assumptions and data sources.

2. Research Review

2.1 NutriScore Lessons

NutriScore demonstrates the usefulness of transparent scoring. Consumers understand a points-based system that balances positives and negatives. NutriScore calibrates across categories to ensure comparability. Similarly, ARS calibrates frequency scores within categories so a cereal and a beverage can be fairly assessed.

2.2 Consumer Apps

Apps like Yuka illustrate that consumers prefer simple, digestible outputs backed by strong research. Yuka maps additives into four levels of risk. ARS follows this pattern by mapping numeric scores into Low, Moderate, High, and Very High, supplemented by short explanations. This improves trust and adoption.

2.3 Medical Risk Modeling

Medical risk modeling often uses severity versus likelihood matrices. ARS extends this model with a third axis: cumulative exposure. This ensures that both short-term acute harms and long-term risks are visible. The weights reflect best practice, giving primacy to severity while integrating frequency and exposure.

3. Criteria and Scales

Three criteria form the backbone of ARS.

Criterion	Definition	Scale (1–5)
Symptom Severity (S)	Clinical seriousness of adverse effects at typical intake levels.	1=negligible; 5=life-threatening

Frequency (F)	Prevalence across foods; proxy for exposure likelihood.	1=rare; 5=very common
Cumulative Exposure (E)	Long-term risk from repeated intake.	1=low concern; 5=high concern

These three criteria are combined using weights to generate an additive's ARS.

4. Scoring Model and Rules

4.1 Formula

The per-additive score uses the weighted formula: ARS_additive = 0.5×S + 0.3×F + 0.2×E. This reflects clinical priority while recognising real-world exposure patterns.

4.2 Aggregation

For products containing multiple additives, three methods are available: (a) Max-risk (highest additive score), (b) Weighted average, and (c) Hybrid (average but no lower than 80% of the max). The hybrid method balances fairness with protection.

4.3 Red Flags and Bands

Red-flag additives automatically trigger at least High risk. Risk bands are defined as: Low (1–1.9), Moderate (2–2.9), High (3–3.9), Very High (≥4). These bands are clear and consumer-friendly.

5. Worked Example

The product Nestlé Multigrain with Banana & Apple contains E304/E304i (ascorbyl palmitate) and E322/E322i (soy lecithin). Allergens listed include gluten, soy, and banana.

E304: Severity=2, Frequency=4, Exposure=2 → ARS=2.6 (Moderate).

E322: Severity=3, Frequency=5, Exposure=3 → ARS=3.8 (High).

Product ARS: Max=3.8 (High), Average=3.2 (High). Soy lecithin drives the risk due to prevalence and allergenic nature.

6. Data and Engineering Design

A registry of additives is essential, containing severity, exposure, ADI, frequency, and red-flag status. Product ingestion processes normalise additive tags, deduplicate variants, and link allergens. The scoring service exposes endpoints for computation, allowing mobile apps to request per-additive and product-level ARS. Cohort profiles ensure that vulnerable users get personalised results. Explainability ensures each score is justified with a short rationale.

The design is modular, enabling updates as new toxicological evidence emerges. It is versioned so users and regulators can trace changes.

7. Validation and Calibration

Validation occurs through expert review, sensitivity analysis, and user testing. Expert panels compare ARS outputs with expected judgments. Sensitivity analysis explores the impact of weight changes. Back-testing against known risky products ensures credibility. User studies test comprehension and trust. Threshold calibration aligns ARS bands to category percentiles, ensuring comparability.

8. Handling Constraints

Label data rarely provide additive concentrations, but ARS mitigates this through frequency and precautionary exposure scoring. Where quantitative data is available, scores are updated accordingly. Stacked exposures (e.g., multiple emulsifiers) add incremental penalties. Regional regulations are tracked, triggering rescoring when changes occur.

A governance process ensures transparency. Every change in scoring or classification is documented with sources, version numbers, and rationale.

9. Roadmap and Deliverables

Phase 1: Build registry, frequency estimator, and API endpoint.
Phase 2: Validate with experts, test cohorts, and adjust thresholds.
Phase 3: Continuous updates, concentration integration, and governance reporting.
Deliverables: (a) this submission, (b) additive registry seed file, (c) scoring service implementation, (d) validation reports, (e) UX rationale library.

This roadmap ensures both immediate functionality and long-term sustainability.

Conclusion

The ARS integrates severity, prevalence, and chronic risk into one composite score. It is transparent, consumer-friendly, and grounded in research. The Nestlé cereal example demonstrates its application, showing that soy lecithin dominates the risk profile. ARS is adaptable, defensible, and scalable.

References

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